

Ardeshir Aidun

Los Gatos, CA | addieaidun@gmail.com | (408) 656-9422 | linkedin.com/in/ardeshir-aidun | github.com/Ardeshir-Aidun

PROFESSIONAL SUMMARY

Electrical Engineering student at UC Santa Barbara with hands-on experience in embedded systems, circuit prototyping, bench testing, and data analysis. Skilled in microcontroller programming, wireless protocols, and hardware debugging, with a track record of designing and validating real hardware systems from concept through testing.

EDUCATION

University of California, Santa Barbara

Bachelor of Science, Electrical Engineering. GPA: 3.72.

Expected June 2028

Relevant Coursework: Digital Design, Analog and Digital Circuits Fundamentals, Signal Analysis, Robotics: Control, Quantum Photonics, Research Process for Engineers

EXPERIENCE

Member

Oct. 2024 – Present

IEEE - UCSB Chapter

Goleta, CA

- Applied soldering techniques and PCB design concepts through hands-on technical workshops
- Assembled and troubleshot circuits, gaining practical experience in component assembly and prototyping

Guest Advocate

June 2025 – June 2026

Target

San Jose, CA

- Coordinated order pickups and curbside deliveries, resolving fulfillment and self-checkout issues

Leader

Aug. 2022 – May 2024

Link Crew

Los Gatos, CA

- Guided incoming students through orientation and campus acclimation activities
- Monitored student progress monthly and provided personal follow-up for those facing academic or social challenges

PROJECTS

ESP32 IoT Mesh Network: Wireless Telemetry and Rollback-Safe OTA Updates

June 2026 – July 2026

- Designed and built a 3-node wireless mesh network using ESP32 microcontrollers, with sensor nodes reporting live temperature and humidity data to a central gateway over ESP-NOW
- Architected a wireless OTA firmware update system using WiFi and HTTP, leveraging the ESP32's dual-partition bootloader to safely stage and verify updates without disrupting the running device
- Validated fault-tolerant recovery by deliberately deploying broken firmware and confirming automatic rollback to the last known-good version with zero manual intervention

Ultrasonic Sensor Accuracy and Noise Characterization

May 2026 – June 2026

- Designed and executed a bench test protocol to characterize accuracy, noise, and drift of an HC-SR04 sensor across a 5-30 cm range using an Arduino Uno
- Built a Python data pipeline to log serial sensor data and compute statistical metrics (mean error, standard deviation, drift rate), visualized with matplotlib
- Identified a distance-dependent measurement bias and noise discrepancy between test protocols, documenting findings in a written report

Automatic Night Light

April 2026 – April 2026

- Designed and built an automated nightlight prototype using an Arduino, a photoresistor (LDR), and an LED circuit
- Programmed real-time ambient light monitoring in C++ using analog input to dynamically control LED output states
- Debugged hardware connections and serial communication errors using the Serial Monitor to optimize system performance

Bike Indicator Lights

Oct. 2024 – Dec. 2024

- Designed and built a custom microcontroller system using an Arduino Uno and push buttons to control left, right, and hazard light functions on an LED strip
- Programmed custom control logic for directional signaling and emergency flashers to improve rider visibility

SKILLS

Hardware: Microcontrollers, Circuit Design, PCB Design, SystemVerilog, Bench Testing, Oscilloscope, Soldering, Prototyping, LTspice

Software and Programming: C++, Python, MATLAB, Simulink, Git/GitHub, Technical Documentation

Communication Protocols: UART, ESP-NOW, WiFi/HTTP

Technical Concepts: Digital Design, Signal Analysis, Boolean Algebra